MT1011 Engineering Materials

Introduction-II

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September 23, 2026





- Solid materials are grouped into three basic categories:
  1. Metals
  2. Ceramics
  3. Polymers
- **Composites** can be considered a fourth category.
- Composites are made of combinations of two or more different materials.
- **Advanced materials** is another class of materials which are developed to be used in high-technology applications.

Material classification

Metals

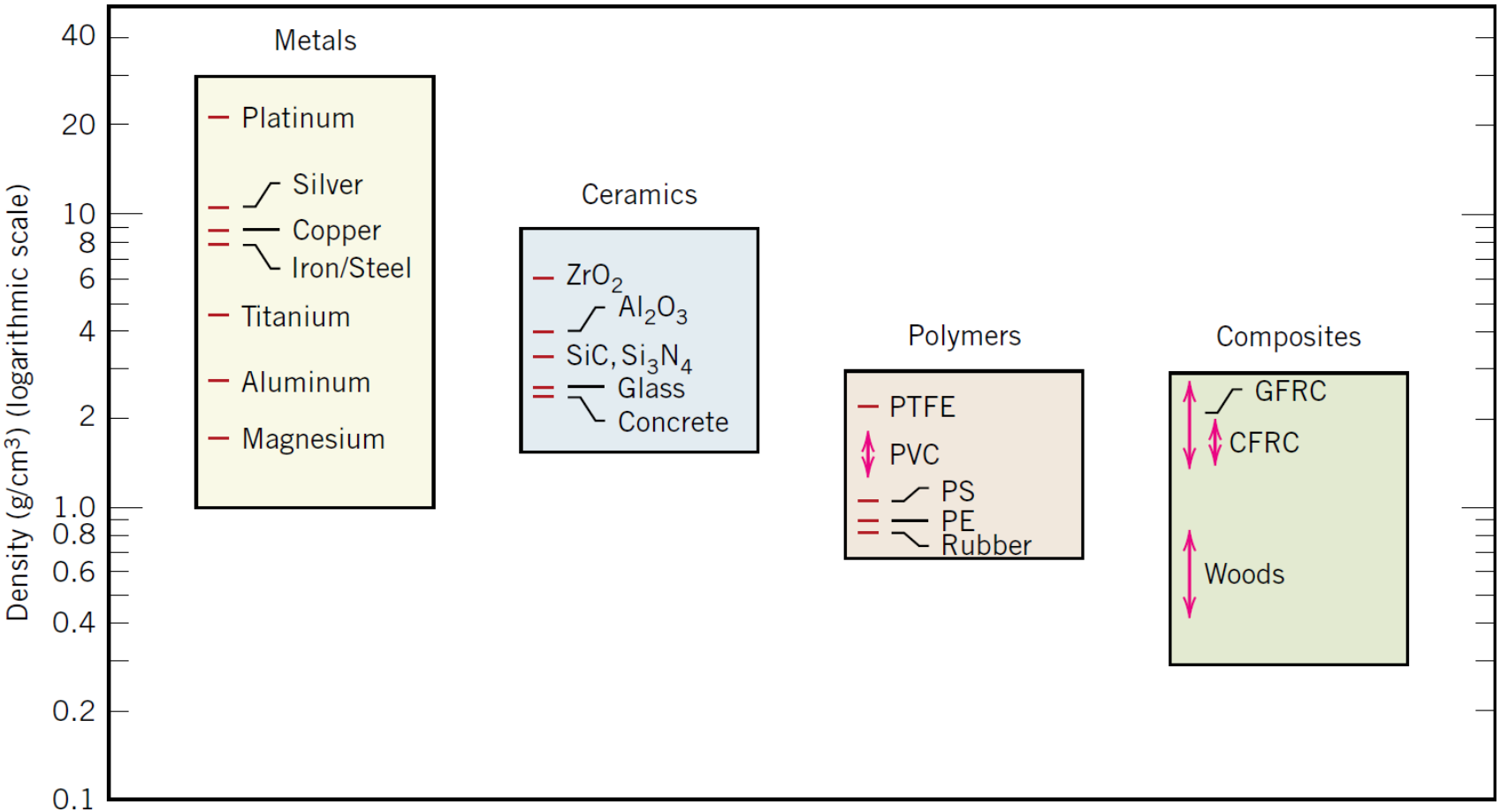


Figure: Room temperature density of selected metallic, ceramic, polymeric, and composite materials.

Picture credit: Callister Jr. and Rethswich (2014)

Material classification

Metals

- Metals are made of one or more metallic elements such as Fe, Ni, Cu, Al, Ti and Au.
- They often also have small amounts of nonmetallic elements such as C, N, Si and O. oxygen.
- Mixtures of metal atoms are called alloys.
- Atoms in metals and alloys are arranged in a very orderly manner in a crystal structure.
- They are therefore relatively more dense than ceramics and polymers.



Material classification

Metals

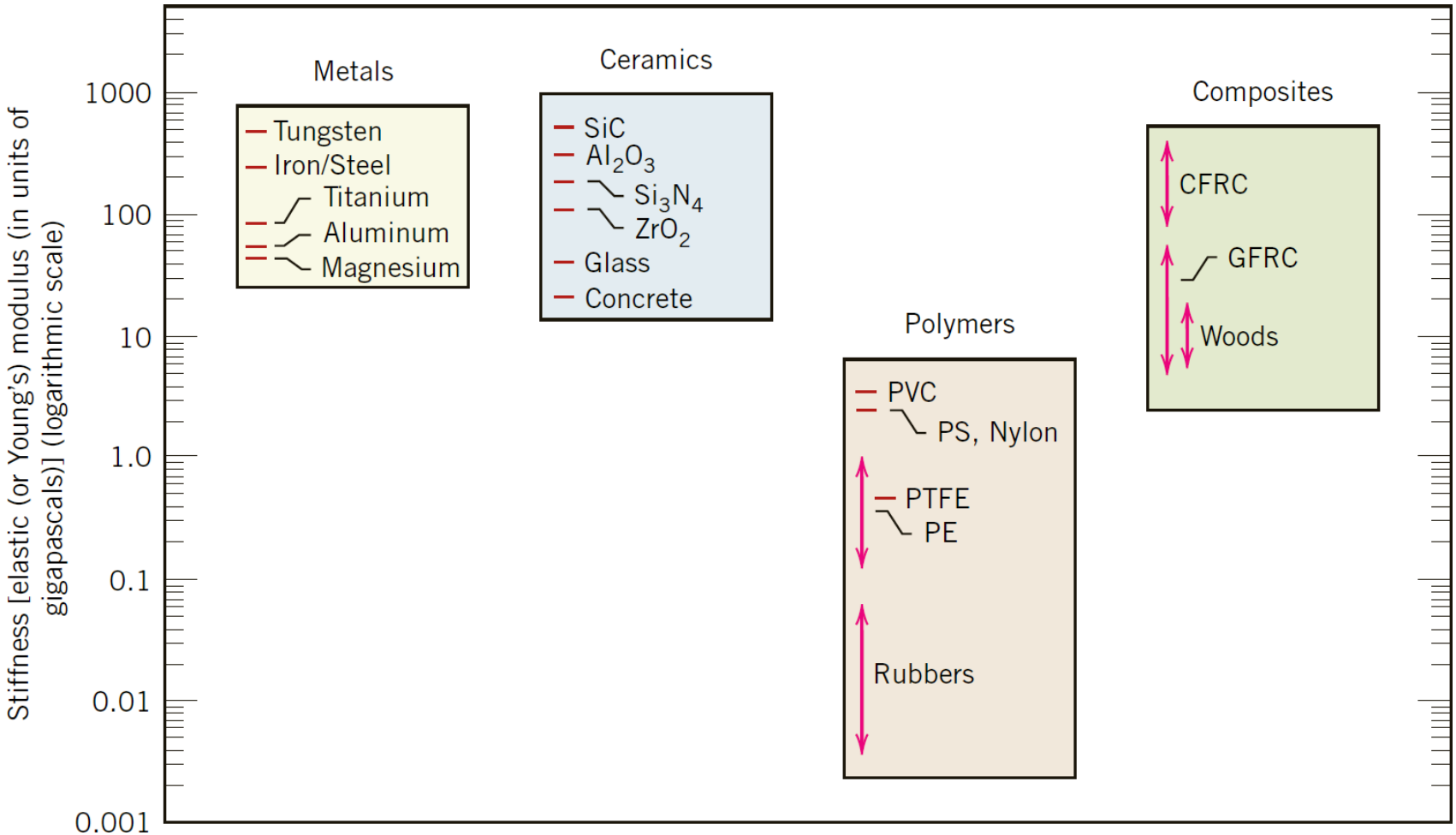


Figure: Stiffness (elastic modulus) of selected metallic, ceramic, polymeric and composite materials. Picture credit: Callister Jr. and Rethswich (2014)

Material classification

Metals

- Metals are relatively more stiff and stronger than ceramics and polymers.
- Metals are also ductile (capable of large amounts of deformation without fracture).
- They are resistant to fracturing.
- Therefore, metals are widely used in structural applications.

Material classification

Metals



Figure: Familiar objects made from metals and alloys.Picture credit: Callister Jr. and Rethswich (2014)

Material classification

Metals

- Metals also have large numbers of delocalized electrons.
- These electrons are not bound to particular atoms and hence very mobile.
- The high electrical and thermal conductivity, and visible light reflectivity (shininess) are their results.
- Some metals additionally show desirable magnetic characteristics (ferromagnetism in Fe, Ni and Co).





Figure: Common objects made of ceramic materials.

- Ceramics can be divided as traditional and technical (or engineering ceramics)
- Traditional ceramics are those composed of clay minerals (e.g. porcelain and terra-cotta), as well as cement and glass.
- Technical ceramics are generally compounds between metallic and nonmetallic elements.
- They are most frequently oxides, nitrides, and carbides.



Material classification

Ceramics

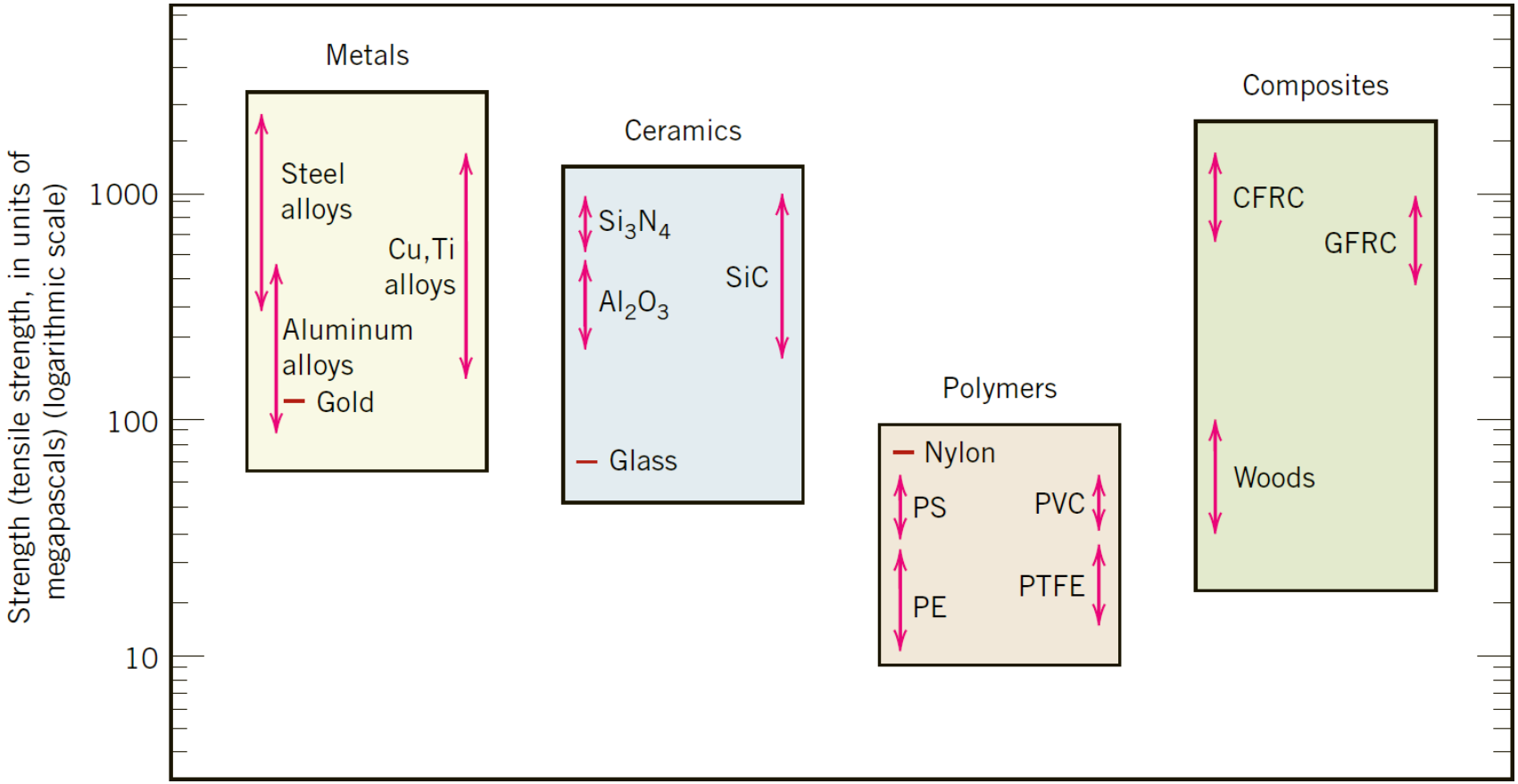


Figure: Tensile strength for various metals, ceramics, polymers, and composite materials.

Picture credit: Callister Jr. and Rethswich (2014)

- Common ceramic materials include:
  - Aluminum oxide (or alumina, Al_2O_3)
  - Silicon dioxide (or silica, SiO_2)
  - Silicon carbide (SiC)
  - Silicon nitride (Si_3N_4)
- Ceramic materials are relatively stiff and strong.
- Stiffnesses and strengths are comparable to those of the metals
- They are typically very hard.
- Historically, ceramics have exhibited extreme brittleness (lack of ductility)
- They are therefore highly susceptible to fracture.



Material classification

Ceramics

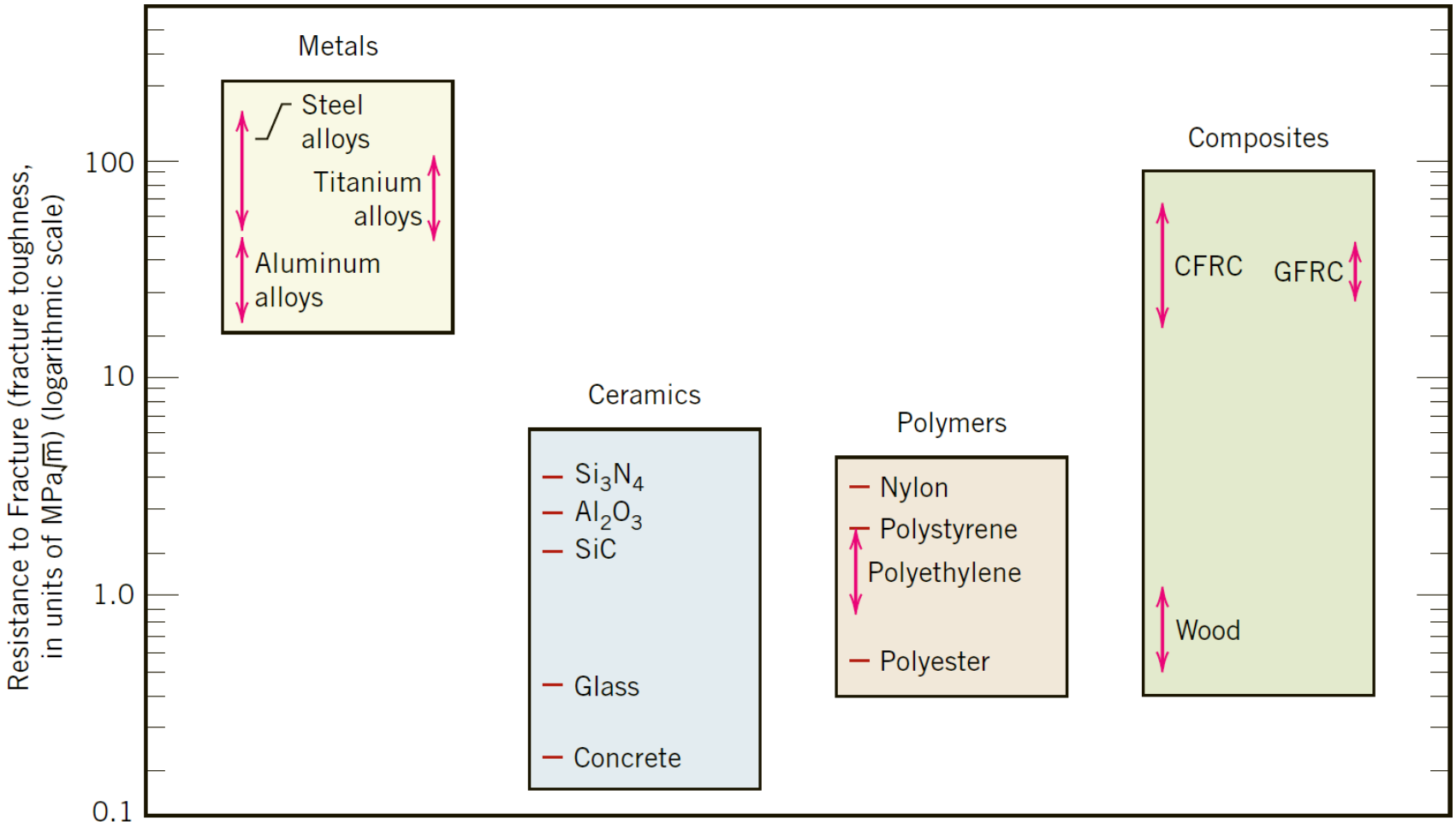


Figure: Fracture toughness of various metals, ceramics, polymers, and composite materials.

Picture credit: Callister Jr. and Rethswich (2014)

- However, newer ceramics are being engineered to have improved resistance to fracture.
- These materials are used for cookware, cutlery, and even automobile engine parts.
- They are typically insulative to heat and electricity (Electrical and thermal conductivity is low).
- They are also more resistant to high temperatures and harsh environments.



Material classification

Ceramics

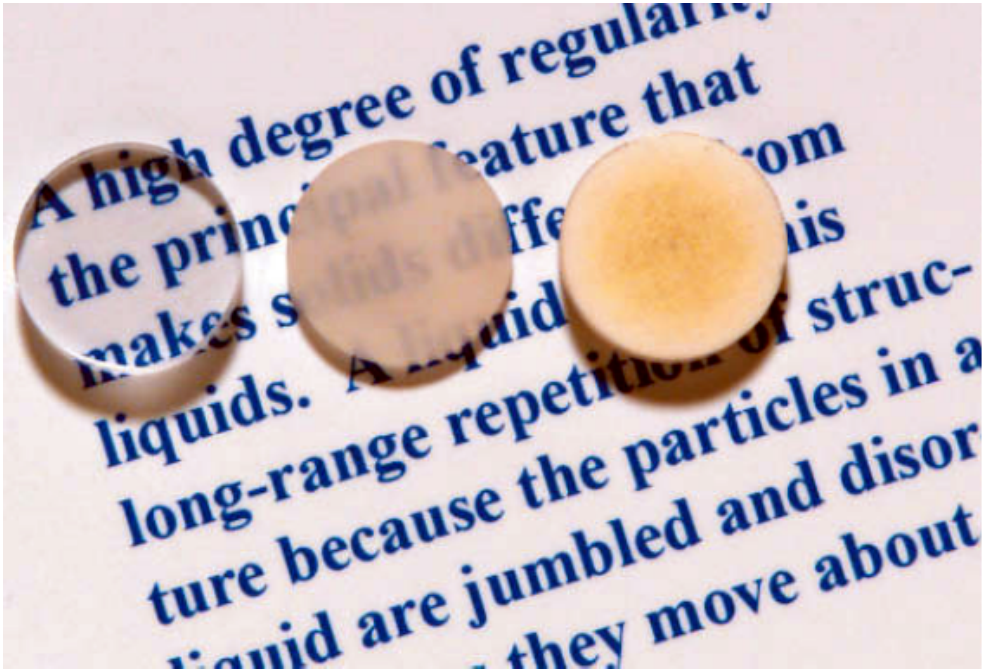


Figure: Different optical properties of alumina Al_2O_3 obtained from different processing routes.

Picture credit: Callister Jr. and Rethswich (2014)

- Ceramics may be transparent, translucent, or opaque.
- Some of the oxide ceramics (e.g. Fe_3O_4) show magnetic behaviour.

Material classification

Polymers



Figure: Objects made from polymeric materials.

Material classification

Polymers

- Polymer class materials include plastic and rubber materials.
- Many of them are organic compounds chemically based on C, H, and other non-metallic elements such as O, N and Si.
- Polymers have very large molecular structures.
- Most have a chainlike structure with a C backbone.
- Some common polymers are polyethylene (PE), nylon, poly(vinyl chloride) (PVC), polycarbonate (PC), polystyrene (PS), and silicone rubber.



Material classification

Polymers



Figure: Objects made from polymeric materials.

Material classification

Polymers

- Polymers generally have low densities.
- Polymers are not as stiff or strong as ceramics or metals.
- Many polymers are very ductile and pliable.
- Therefore, they are easily formed into complex shapes.
- Generally, they are chemically inert.
- One major drawback is their tendency to soften and/or decompose at modest temperatures.
- Polymers have low electrical conductivities and are non-magnetic.

